AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-28 | Entry window 28 October 2026 (full day) | 22 records screened in scope

22 RECORDS IN SCOPE (96 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

8 EFFECT ESTIMATES WITH A CI

2 TIER-A RECORDS

7 RECORDS CARRIED ON METADATA ALONE

Signal of the day

West Oakland: a dense black-carbon network plus Bayesian inversion raises the BC inventory by a third and quadruples the exposure share of neighbourhood and port sources.

Mobile and fixed-site BC measurements (6.7 sites km^{-2}) were assimilated into hourly 150 m emission maps. The posterior added ∼33 % to bottom-up BC emissions and moved the population-weighted exposure share of local and port-related sources from 5 to 20 %, concentrated in low-income blocks beside freight corridors. A thinned network of ∼10 fixed monitors (∼1 km^{-2}) plus time-averaged mobile maps gave similar answers. **Where it is weak:** the posterior inherits the transport kernel and prior error covariance; "unaccounted" sources are inferred, not inspected, and this is one community. **Why it matters here:** it is the clearest demonstration yet that a sensor network's value is in inversion, not in concentration maps — and it states the density needed.

What changed today

- **Urban firestorm smoke is a metal plume.** Eaton fire, Los Angeles: lead >50-fold, halogens 7-fold, arsenic 3-fold above wildfire smoke; ∼7 million residents above 50 ng m^{-3} 24-h lead. Optical sensors see the mass, not the toxicity.
- **Low-cost sensors as the sole exposure metric.** Kinshasa LBW study used calibrated PurpleAir PA-II-SD (0.52×raw + 3.2 vs BAM): area $PM_{2.5}$ >75 $\mu g\,m^{-3}$ aOR 1.18 (0.59–2.36) — but sensors ran 48–72 h in 2022 for births in 2021.
- **Component scaling artefact.** China 123-city cohort: BC HR 1.381 per 1 $\mu g\,m^{-3}$ vs $PM_{2.5}$ 1.027 — a per-unit comparison across ranges that differ tenfold.
- **Paediatric asthma, Louisiana.** 28,499 hospitalisations: $PM_{2.5}$ lag 0–3 OR 1.079 (1.052–1.107), robust to co-pollutants; no significant modification by race or Medicaid.
- **Cross-day duplicates.** 14 records in the 28 October harvest had already entered the 27 October pool via Europe PMC/PubMed lag and were issued there; one record, one issue.
- **Metadata-only share 32 %.** Seven day-0 Elsevier/T&F deposits with no abstract anywhere; abstract retry (EPMC → Semantic Scholar) recovered none.

Window continuity

The 22 October issue closed with `last_entry_date` at 22 October. Scheduled runs for 23–29 October did not complete: a run at 00:21 UTC on 30 October harvested 23–28 October but stopped before screening. This issue was built on the **30 October 2026 14:20 UTC (09:20 CDT) recovery run**, which backfills 23–29 October day by day, each with its own window and issue. This issue collects **28 October in full**, continuous with the 27 October issue.

Corpus analytics

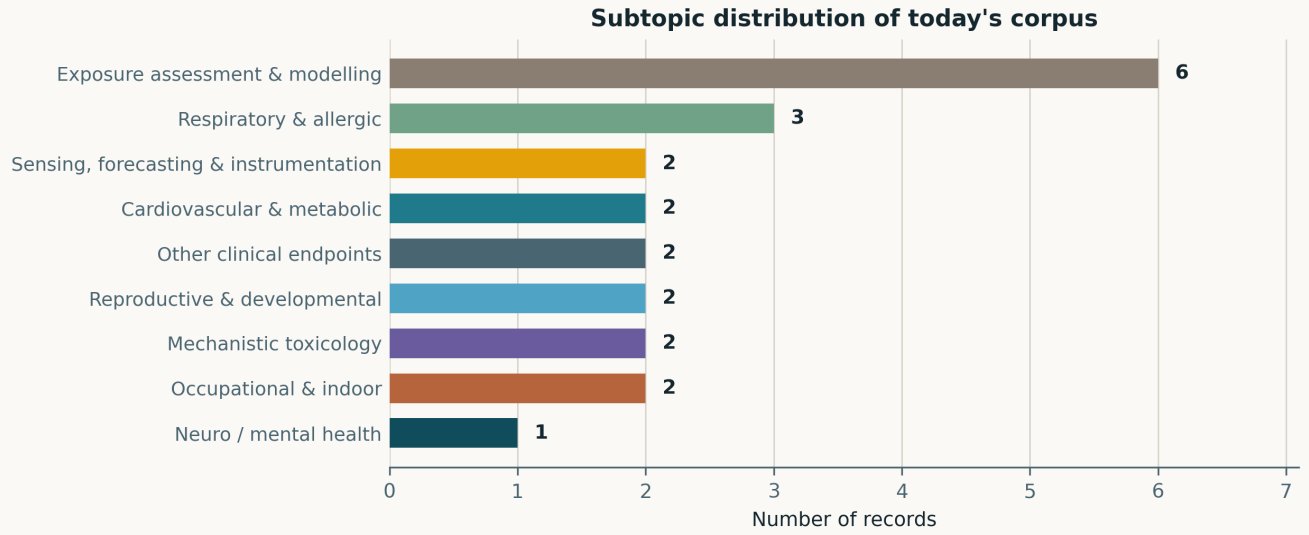


Fig. Subtopic assignment of the 22 in-scope records across nine bands. Exposure assessment leads with six (urban-fire metals, fireworks, PM–O_3 coupling and three metadata-only records); respiratory three; burden/policy is empty.

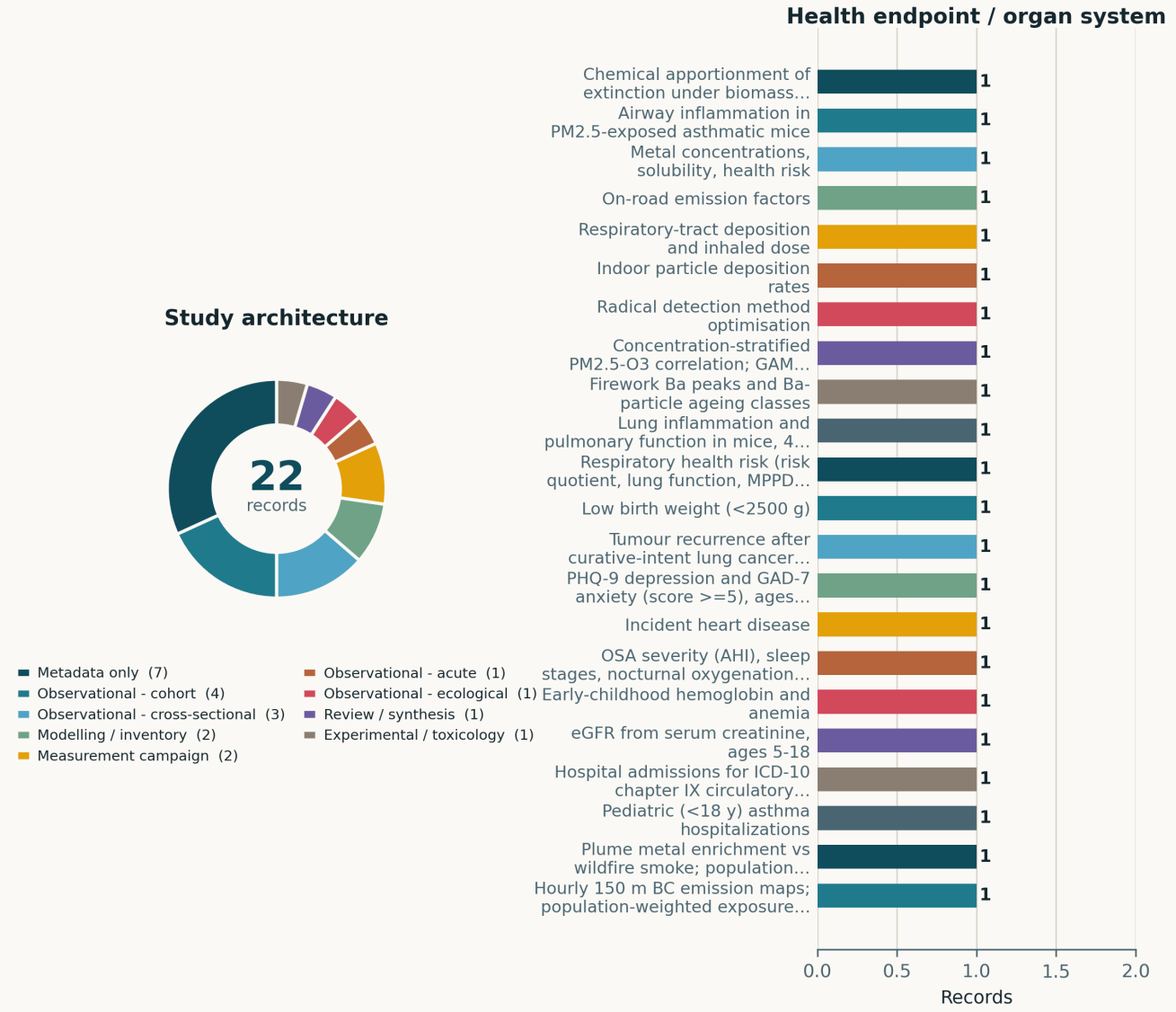


Fig. Left — architecture: cohorts, cross-sectional surveys, field measurements, one inversion method, one case-crossover, one ecological panel; **metadata-only 7 (32 %)**. Right — endpoints run from LBW, childhood haemoglobin and eGFR to asthma admissions, OSA, heart disease and cancer recurrence.

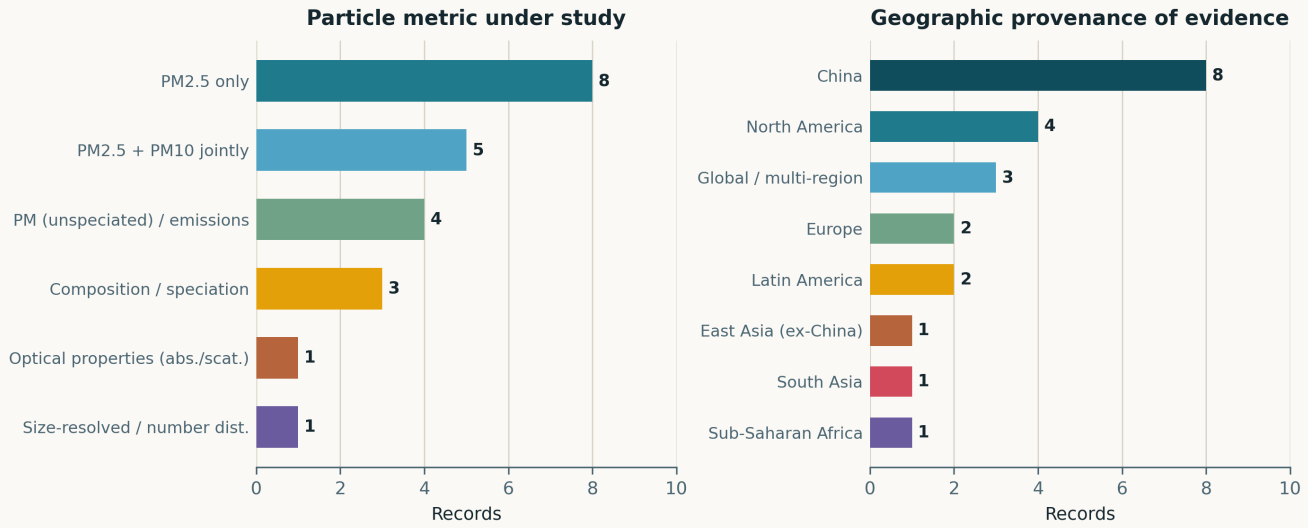


Fig. Left — particle metric: $PM_{2.5}$-only dominates; BC and speciated metals carry the two tier-A records. Right — geography: China leads; USA second (Oakland, Los Angeles, Louisiana); one each for DRC and Brazil; several metadata-only records carry inferred locations.

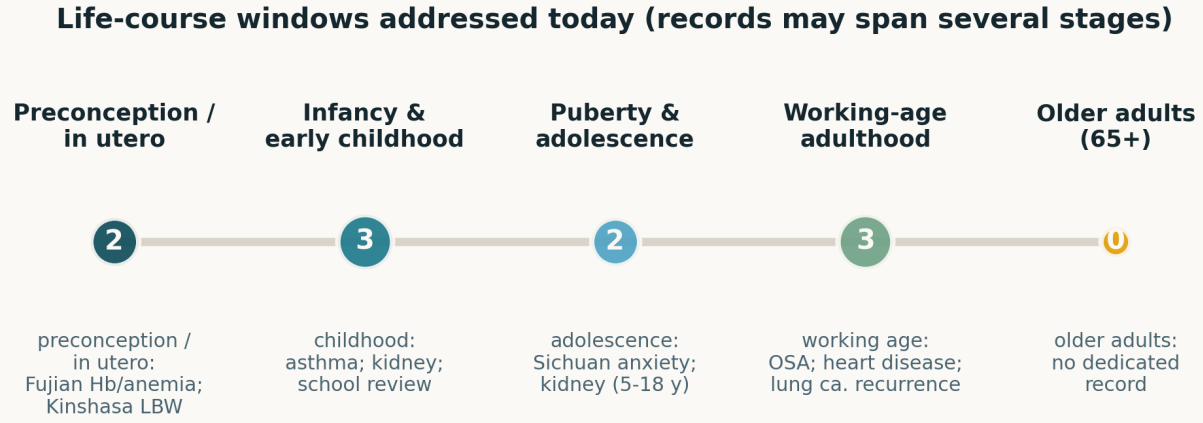


Fig. Life-course windows: in utero two (Fujian haemoglobin/anaemia, Kinshasa LBW); childhood three; adolescence two; working age three; no dedicated older-adult record.

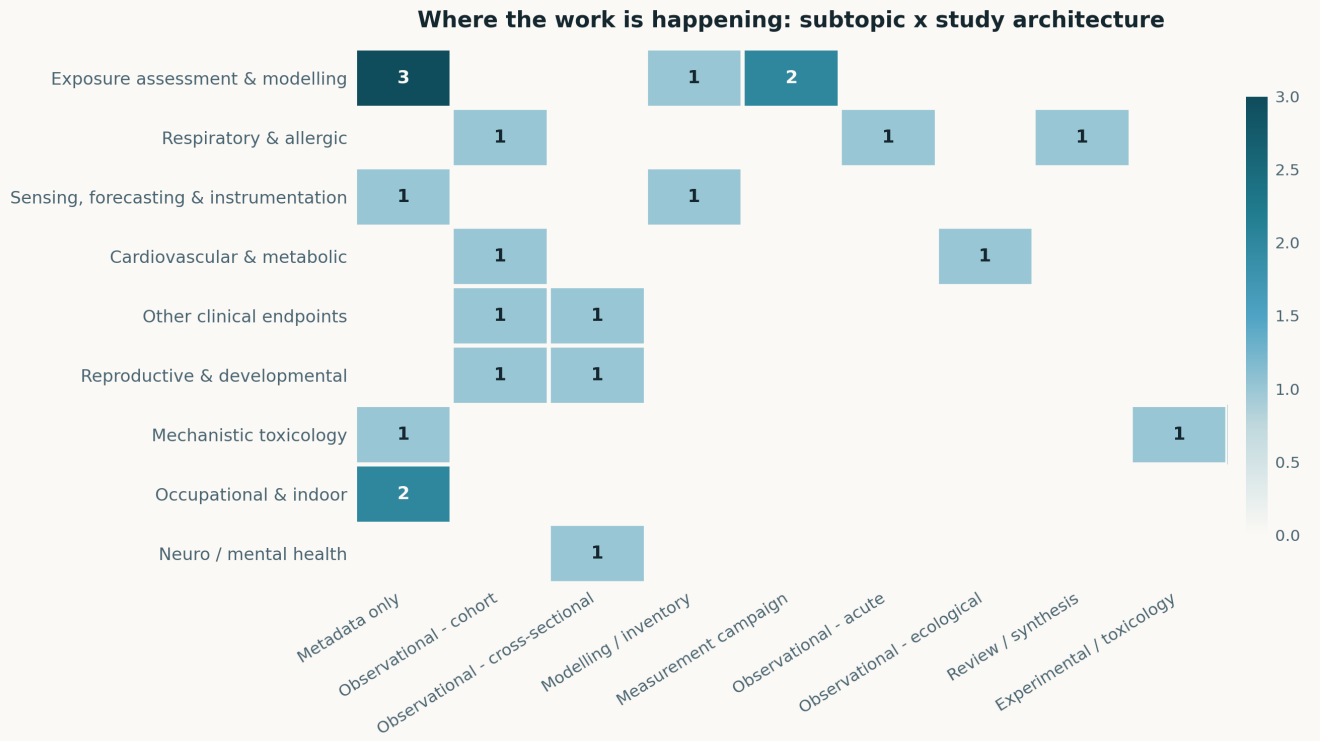


Fig. Subtopic × study architecture. Exposure assessment spreads over measurement and metadata-only cells; the health bands sit in cohorts and cross-sectional surveys; sensing holds the inversion method and one metadata-only metrology record.

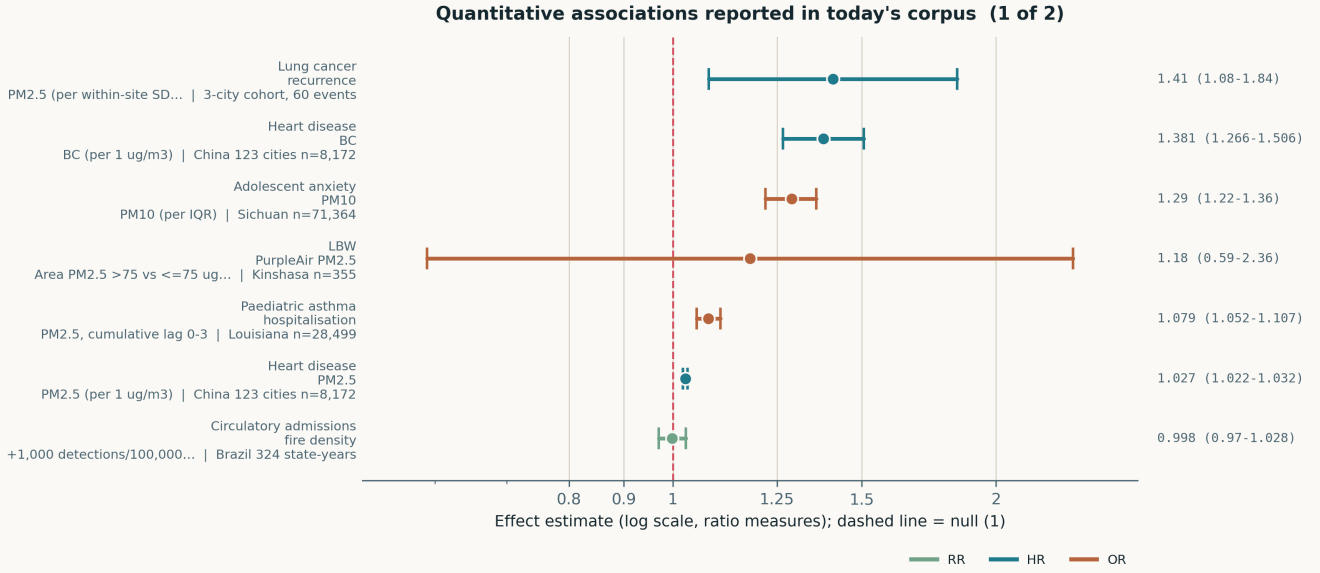


Fig. Ratio scale, seven estimates from six studies: Louisiana paediatric asthma ($PM_{2.5}$ lag 0–3); China 123-city heart disease per 1 $\mu g\,m^{-3}$ $PM_{2.5}$ and BC; lung cancer recurrence per within-site SD; Sichuan adolescent anxiety per IQR PM_{10}; Kinshasa LBW (area $PM_{2.5}$ >75 $\mu g\,m^{-3}$); Brazil circulatory admissions per fire-detection density. Contrasts differ completely. Not poolable.

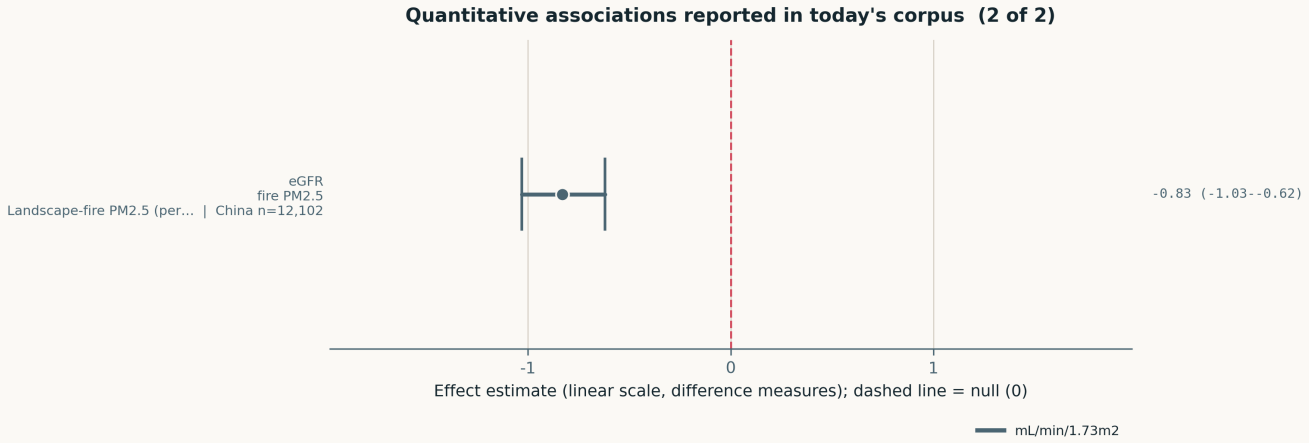


Fig. Difference scale: eGFR (mL min^{-1} 1.73 m^{-2}) per IQR (0.505 $\mu\mathrm{g}\,\mathrm{m}^{-3}$) of landscape-fire $PM_{2.5}$ in 12,102 Chinese children — a large effect for a tiny contrast; read with the confounding caveat.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Wang air pollution adolescent anxiety depression Sichuan cross-sectional 2026 BMC Public Health

Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China

71,364 school students aged 10-19, surveyed over two winters (2022-23, 2023-24). Per IQR, PM_{10} anxiety OR 1.29 (1.22-1.36) and O_3 1.24 (1.19-1.29), with similar depression associations; a weighted combined score rose stepwise. $PM_{2.5}$ and SO_2 were inversely associated. Limitation: exposure was assigned from the IP address of the online survey, i.e. a point-in-time location of the device, and the PM_{10}-positive/$PM_{2.5}$-inverse split from highly correlated pollutants is the signature of collinearity and spatial confounding rather than a fraction-specific effect. Relevance: low - coarse exposure assignment.

doi: 10.1186/s12889-026-28370-5

Cardiovascular & metabolic — 2 records

Andreotti active fire detections circulatory hospital admissions Brazil ecological panel 2026 Cureus

Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study (2013-2024)

27 Brazilian federative units x 12 years (324 unit-years), 13.8 million circulatory admissions and 2.3 million fire detections. Correlations were null (rho -0.03 to 0.01). In negative-binomial models with state fixed effects, year effects, temperature, poverty and primary-care coverage, the rate ratio per additional 1,000 detections/100,000 km2/year was 0.998 (0.970-1.028), null in every subset and with clustered SEs. Limitation: annual state-level fire counts are far removed from short-term smoke $PM_{2.5}$ at residence; smoke from Amazon fires is exported to other states, so the exposure is misplaced as well as aggregated. The authors correctly state this is not evidence of no effect. Relevance: low - illustrates why fire counts cannot substitute for measured PM.

doi: 10.7759/cureus.118676

Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities 2026 BMC Cardiovasc Disord

Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China

8,172 participants from 123 Chinese cities; Cox models with three adjustment levels plus WQS mixtures. Per 1 $\mu\mathrm{g}\,\mathrm{m}^{-3}$ $PM_{2.5}$: HR 1.027 (1.022-1.032); BC had the largest per-unit estimate, HR 1.381 (1.266-1.506). All components showed non-linear exposure-response; southern residents and overweight participants had stronger associations; PAF 2.63%. Limitation: per-$\mu\mathrm{g}\,\mathrm{m}^{-3}$ component HRs are not comparable because component ranges differ by an order of magnitude, and components are highly collinear, so 'BC largest' is a scaling artefact unless expressed per IQR. Heart disease is likely self-reported (cohort not named in the abstract). Relevance: medium - again points to absorption (BC) as the health-relevant channel.

doi: 10.1186/s12872-026-06395-3

Respiratory & allergic — 3 records

Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups 2026 Environ Epidemiol

Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana

28,499 asthma hospitalizations in children in Louisiana 2010-2019, time-stratified case-crossover with DLNM to lag 3. Cumulative lag 0-3: $PM_{2.5}$ OR 1.079 (1.052-1.107), O_3 1.082 (1.050-1.116), NO_2 1.071 (1.034-1.111), with increments as defined in the paper. In multipollutant models O_3 and NO_2 attenuated while $PM_{2.5}$ persisted. Effect modification by race and Medicaid status was not significant, although lag patterns differed (lags 1-2 Black children, lag 3 White children). Limitation: exposure assigned from area-level estimates in a state with sparse monitoring outside the industrial corridor, so differential misclassification by race/rurality could mask modification; NO_2 covers only 2010-2016. Relevance: medium - higher-burden groups sit where reference coverage is thinnest.

doi: 10.1097/ee9.0000000000000544

Wang air pollutants sleep apnea polysomnography architecture Shenzhen 2026 Environ Res

Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea (OSA)

Retrospective polysomnography study of 4,049 adults in Shenzhen. Over a 0-180 day lag window, $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO and diurnal temperature range were associated with greater OSA severity. $PM_{2.5}$ and PM_{10} were associated with 1.71% and 1.32% of total sleep time more N1 and 1.51% and 1.32% less N3 sleep (increments as reported). O_3 and DTR showed the clearest associations with oxygenation; $PM_{2.5}$ had a rising exposure-response. Limitation: the lag window was chosen post hoc, which inflates false-positive risk across many pollutant-outcome pairs; a clinic-referred sample means referral itself may track season and pollution. CIs are not given in the abstract. Relevance: low - city-level exposure.

doi: 10.1016/j.envres.2026.126044

Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review 2026 Research Square (preprint)

Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review

PRISMA review (PROSPERO CRD420261282406) of 12 studies published 2015-2025 on school $PM_{2.5}$ and PM_{10}, combining environmental health risk assessment, epidemiology, MPPD dosimetry and STELLA modelling. Several studies reported risk quotients >1; one Bandung study found inverse $PM_{2.5}$-lung function associations; dosimetry indicated higher airway deposition in children. Limitation: 12 heterogeneous studies with mixed methods and no quantitative synthesis; RQ>1 depends on default intake assumptions rather than measured outcomes. Preprint. Relevance: medium - underlines the need for standardised school-level monitoring that low-cost networks could provide.

doi: 10.21203/rs.3.rs-11094408/v1

Reproductive & developmental — 2 records

Zhu prenatal PM_1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort 2026 Ecotoxicol Environ Saf

Association of prenatal PM_1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows

Fujian Birth Cohort paediatric subcohort, 1,235 mother-child pairs. Per 1 $\mu g\,m^{-3}$ over the whole pregnancy, child hemoglobin fell by 1.02 (0.53-1.52) for PM_1, 0.87 (0.46-1.28) for $PM_{2.5}$ and 0.37 (0.19-0.56) for PM_{10} (units as reported). First-trimester anemia ORs per unit were 1.23, 1.16 and 1.09; DLMs placed the hemoglobin window in early-mid pregnancy and the anemia window in the first trimester. Limitation: per-microgram effects of this size imply implausibly large decrements across the cohort's range; PM_1, $PM_{2.5}$ and PM_{10} are near-collinear satellite products, so the size-fraction ranking is not interpretable. Postnatal exposure and diet are not stated. Relevance: low - satellite-derived exposures only.

doi: 10.1016/j.ecoenv.2026.120980

Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC 2026 J Toxicol

Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022

355 mothers delivering in 2021 at six facilities near the Carrigres quarry. LBW prevalence 12.96% (9.7-16.9). PurpleAir PA-II-SD sensors, corrected with a Kinshasa BAM-derived equation, gave area $PM_{2.5}$ 54.6-102.5 $\mu g\,m^{-3}$. Residence <500 m from the quarry: aOR 4.12 (1.98-8.57); area $PM_{2.5}$ >75 $\mu g\,m^{-3}$: aOR 1.18 (0.59-2.36). Limitation: $PM_{2.5}$ was measured for 48-72 h in 2022 and assigned to births in 2021, i.e. after the pregnancies, at the health-area level, so the null $PM_{2.5}$ result says nothing about gestational exposure; quarry dust is coarse and poorly captured by Plantower optics. Relevance: high - a direct example of low-cost sensors used as the sole exposure metric, and of the temporal mismatch that undermines it.

doi: 10.1155/jt/7953451

Mechanistic toxicology — 2 records

Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride 2026 Environ Res

Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires

Soot (10 fire tests) and smoke PM (2 tests) from lithium-ion battery thermal runaway were size-filtered to <10 um and given to mice (100 ug). Soot had more carbon (37% vs 18%), PM more anions (12% vs 6%), inorganic elements similar (23-28%); particulate fluoride reached ~5% of mass. Only soot altered ventilatory timing, peak expiratory flow and relaxation time; both raised lung inflammation to levels comparable with other combustion sources. Fluoride, Al, Fe, Ni and Li were associated with toxicity. Limitation: a single 100 ug bolus (aspiration/instillation) is not inhalation dosimetry, two smoke tests is a thin basis, and component-toxicity associations across so few samples are exploratory. Relevance: low - toxicology; flags fluoride-bearing PM as a new indoor/urban fire hazard.

doi: 10.1016/j.envres.2026.126023

Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$ 2026 Environ Int

Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$

METADATA ONLY. Title and authors only (Liu, Sun, Wang, Jin et al.); no abstract retrievable at harvest. The title describes a probiotic-derived fermentation supernatant protecting asthmatic mice against $PM_{2.5}$-aggravated airway disease. Dose, model and outcomes cannot be stated. Limitation: unverifiable beyond the title; $PM_{2.5}$ aggravation models usually use instilled boluses far above ambient deposition. Relevance: low.

doi: 10.1016/j.envint.2026.110584

Exposure assessment & modelling 6 records

Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles 2026 Sci Adv

Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm

Hourly airborne trace-element measurements during the January 2025 Eaton fire tracked the transition from structure to wildland burning via the K/Pb ratio. Relative to wildfire smoke, the urban plume was enriched >50-fold in lead, ~7-fold in halogens and ~3-fold in arsenic, attributed to vehicles, electronics and building materials. An estimated 7 million residents (range 4.3-8.0 million) experienced 24-h lead above 50 ng/m3, >80 times background. Limitation: population exposure is extrapolated from a small number of monitoring points through a dispersion estimate, and the 24-h lead threshold has no health-based standard counterpart (NAAQS is a 3-month mean). Relevance: high - optical PM sensors would register the smoke but not its metal toxicity; speciation is needed.

doi: 10.1126/sciadv.aei4455

Lian firework barium aerosols single-particle SPAMS Guangzhou 2026 Atmos Chem Phys

Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring

Online Ba mass monitoring and SPAMS at four Guangzhou sites through firework events. Ba rose 1-3 orders of magnitude above background. The urban-core site under a complete firework ban peaked lower than a suburban site with time- and place-restricted use, despite higher population density - consistent with, but not proof of, a ban effect. Chloride-rich and nitrate-dominated Ba-containing particle classes were interpreted as fresh and aged, tracking chloride displacement and nitrate uptake. Limitation: four sites and a few events; the ban contrast is confounded by local emissions and transport, as the authors acknowledge, and SPAMS is semi-quantitative. Relevance: medium - firework episodes are a known optical-sensor artefact (hygroscopic, metal-rich particles), and this characterises their chemistry.

doi: 10.5194/acp-26-15101-2026

Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025 2026 Atmosphere

Long-Term Trends and Nonlinear $PM_{2.5}$-O_3 Coupling in a Typical Oilfield City of the "2 + 26" Region

Daily $PM_{2.5}$ and MDA8 O_3 in Puyang 2015-2025 with nested GAMs adding meteorology, NO_2, humidity and season interactions. Raw correlation reversed by concentration band: r 0.271 below 35 $\mu g\,m^{-3}$, -0.208 at 35-75 and -0.313 above 75. After controlling for meteorology and NO_2, the conditional partial effect of $PM_{2.5}$ on O_3 increased over most of the range, so the negative raw correlation is largely meteorological. The low-band positive coupling strengthened over time (0.190 to 0.306). Limitation: one city, observational correlations; GAM partial effects do not identify chemical mechanisms (aerosol radiative or heterogeneous HO2 uptake). Relevance: low.

doi: 10.3390/atmos17111057

Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants 2026 Atmos Pollut Res

On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants

METADATA ONLY. Title and authors only (Jabbari-Hichri, Azizi, Guiot, Boreave et al.); no abstract retrievable at harvest. The title describes detailed on-road measurement of particulate and gaseous emissions from light-duty vehicles. Vehicles, instruments and emission factors cannot be stated. Limitation: unverifiable beyond the title; on-road (PEMS) particle number measurements usually exclude volatile and sub-23 nm particles, which matters for comparing with roadside data. Relevance: low.

doi: 10.1016/j.apr.2026.103252

Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk 2026 Atmos Pollut Res

Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment

METADATA ONLY. Title and authors only (Netto, da Silva, Diniz et al.); no abstract retrievable at harvest. The title describes $PM_{2.5}$ metal concentrations, their soluble fractions and risk assessment in urban areas with smelting in Rio de Janeiro state. Concentrations, n and risk estimates cannot be stated. Limitation: unverifiable beyond the title; risk assessments on total metal mass overstate dose unless the bioaccessible (soluble) fraction is used, which the title suggests is addressed. Relevance: low - filter chemistry.

doi: 10.1016/j.apr.2026.103250

Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific 2026 Atmos Res

Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific

METADATA ONLY. Title and authors only (Pani, Lee, Lin); no abstract retrievable at harvest. The title describes apportioning aerosol light extinction by chemical component at a high-altitude remote site in the western North Pacific under transported Southeast Asian biomass-burning smoke. Concentrations and apportionment fractions cannot be stated. Limitation: unverifiable beyond the title; IMPROVE-type mass extinction efficiencies are fixed assumptions that aged smoke violates. Relevance: medium - mass-extinction relations are what optical PM sensors implicitly assume.

doi: 10.1016/j.atmosres.2026.109421

Sensing, forecasting & instrumentation 2 records

Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland 2026 Sci Adv

Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions

Dense mobile and fixed-site BC measurements were assimilated in a Bayesian inversion to produce hourly 150 x 150 m emission maps for West Oakland, a freight-corridor community. The posterior raised total BC emissions by ~33% over the bottom-up inventory and quadrupled the share of population-weighted exposure attributed to neighbourhood and port-related sources (5% to 20%), concentrated in low-income blocks. Sensitivity runs with ~10 fixed monitors (~1 site/km2) plus time-averaged mobile maps recovered similar results. Limitation: the posterior is only as good as the transport kernel and prior error covariance; unpermitted sources are inferred, not verified on the ground, and one community limits transfer. Relevance: high - a direct recipe for turning a sparse low-cost BC/PM network into source-resolved emission corrections.

doi: 10.1126/sciadv.aeb7647

Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor 2026 Aerosol Sci Technol

Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor

METADATA ONLY. Title and authors only (Zhao, Chen, Ding, Zhang et al.); no abstract retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes optimising EPR spin-trapping for radicals generated by $PM_{2.5}$, identifying the time course of adduct formation and the cavity quality factor as critical. Sample type, n and performance metrics cannot be stated. Limitation: unverifiable beyond the title; spin-trap adducts decay on minute timescales, so inter-lab oxidative-potential comparisons hinge on exactly these timing choices. Relevance: low - offline oxidative-potential metrology.

doi: 10.1080/02786826.2026.2750406

Occupational & indoor 2 records

Kulmala determination deposition rates indoor particle dynamics 2026 Aerosol Sci Technol

Determination of deposition rates from indoor particle dynamics

METADATA ONLY. Title and authors only (Kulmala I., Siilin, Hakala, Ehder-Gahm); no abstract retrievable at harvest. The title describes deriving particle deposition rates from indoor concentration dynamics. Method, size range and validation cannot be stated. Limitation: unverifiable beyond the title; decay-based deposition estimates are confounded by air-exchange and mixing assumptions unless these are measured independently. Relevance: medium - deposition rates are an input for indoor/outdoor ratio models used to interpret paired indoor and outdoor low-cost sensors.

doi: 10.1080/02786826.2026.2754776

Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities 2026 Atmos Environ

Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities

METADATA ONLY. Title and authors only (Ng, Monalisa, Balachandran, Hoque et al.); day-0 Elsevier deposit with no abstract at harvest. The title describes household PM measurement in biomass-burning homes in rural indigenous communities with respiratory deposition and inhaled-dose modelling. Instruments, n and concentrations cannot be stated. Limitation: unverifiable beyond the title; deposition models (e.g. MPPD) are sensitive to the assumed size distribution, which optical monitors in smoke-laden kitchens report poorly. Relevance: medium - household biomass smoke is a key use case for low-cost PM monitors.

doi: 10.1016/j.atmosenv.2026.122448

Other clinical endpoints 2 records

Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China 2026 Environ Res

Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China

China National Nutrition and Health Surveillance of Children and Lactating Women 2016-2018, n=12,102 aged 5-18. Per IQR (0.505 $\mu g\,m^{-3}$) of fire-sourced $PM_{2.5}$: eGFR -0.83 mL/min/1.73 m2 (-1.03 to -0.62); middle-school students -1.46 (-1.83 to -1.09) vs primary -0.59 (-0.84 to -0.34). Associations were stronger with low NDVI and low vitamin or carbohydrate intake. Limitation: an IQR of 0.5 $\mu g\,m^{-3}$ is a tiny contrast for an effect this size, which points to residual spatial confounding (fire-PM tracks rural, agricultural regions); creatinine-based eGFR in children depends on muscle mass and hydration, which also vary by region. Cross-sectional. Relevance: low - source-specific PM from a chemical-transport model, no sensing.

doi: 10.1016/j.envres.2026.125978

Hu air pollution built environment lung cancer recurrence after surgery China 2026 Research Square (preprint)

Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities

542 adults after curative resection at three tertiary hospitals (2019-2026); 60 recurrences, median follow-up 27.3 months. Site-stratified meta-analysis, per within-site SD of first-year post-diagnosis exposure: $PM_{2.5}$ HR 1.41 (1.08-1.84), PM_{10} 1.38 (1.10-1.73), NO_2 1.67, CO 1.51, SO_2 1.59; O_3 null. A co-exposure component gave HR 1.51 (1.20-1.89); green-space diversity HR 0.59/SD. Limitation: 60 events across six pollutants, mixtures and built-environment terms is badly underpowered for adjustment; within-site SD contrasts are small and residence tracks stage, histology and access to follow-up imaging. Preprint. Relevance: low.

doi: 10.21203/rs.3.rs-11311605/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Wang air pollution adolescent anxiety depression Sichuan cross-sectional	Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China	*BMC Public Health*	School-based cross-sectional	PM_{10}, $PM_{2.5}$, O3, CO, SO2, NO2 assigned via IP-geolocated survey location; winter periods	China (Sichuan)
Cardiovascular & metabolic					
Andreotti active fire detections circulatory hospital admissions Brazil ecological panel	Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study	*Cureus*	Ecological panel, fixed effects	Annual satellite active-fire detection density (INPE) as smoke proxy	Brazil (27 federative units)
Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities	Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China	*BMC Cardiovasc Disord*	Prospective cohort	$PM_{2.5}$ and components (SO4, NO3, NH4, OM, BC), gridded	China (123 cities)
Reproductive & developmental					
Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC	Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022	*J Toxicol*	Cross-sectional survey	Calibrated PurpleAir PA-II-SD $PM_{2.5}$ (48-72 h per health area; 0.52 x raw + 3.2 vs BAM)	Democratic Republic of the Congo (Kinshasa)
Zhu prenatal PM1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort	Association of prenatal PM1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows	*Ecotoxicol Environ Saf*	Prospective birth cohort	Trimester-specific PM1, $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants)	China (Fujian)
Respiratory & allergic					
Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review	Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review	*Research Square (preprint)*	Systematic review (PRISMA)	Indoor and outdoor school $PM_{2.5}$ and PM_{10}	Global / multi-region
Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups	Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana	*Environ Epidemiol*	Time-stratified case-crossover	Daily $PM_{2.5}$, O3 and NO2 (NO2 2010-2016 only)	USA (Louisiana)
Wang air pollutants sleep apnea polysomnography architecture Shenzhen	Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea	*Environ Res*	Retrospective cohort	$PM_{2.5}$, PM_{10}, $PM_{2.5}$ inorganic components, NO2, SO2, CO, O3; lag 0-180 d	China (Shenzhen)
Mechanistic toxicology					
Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$	Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$	*Environ Int*	Metadata only (no abstract)	$PM_{2.5}$ exposure in an asthmatic mouse model	Laboratory (not stated)
Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride	Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires	*Environ Res*	Animal exposure study	LIB thermal-runaway soot and smoke PM (<10 um), chemical composition	USA (fire test facility; mouse exposures)
Occupational & indoor					
Kulmala determination deposition rates indoor particle dynamics	Determination of deposition rates from indoor particle dynamics	*Aerosol Sci Technol*	Metadata only (no abstract)	Indoor particle concentration decay (size-resolved, presumed)	Finland (inferred from author affiliation; not stated in record)
Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities	Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities	*Atmos Environ*	Metadata only (no abstract)	Household PM from biomass burning; respiratory deposition / inhaled dose	India (inferred from author affiliation; not stated in record)
Sensing, forecasting & instrumentation					
Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland	Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions	*Sci Adv*	Method / algorithm	Black carbon from dense fixed-site network (6.7 sites/km2) plus mobile mapping	USA (West Oakland, California)

First author	Focus	Journal	Design	Metric	Region
Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor	Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor	*Aerosol Sci Technol*	Metadata only (no abstract)	$PM_{2.5}$-induced radicals (EPR spin trapping)	Laboratory (not stated)
Exposure assessment & modelling					
Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles	Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm	*Sci Adv*	Field measurement	Real-time airborne trace elements (XRF) in PM during the Eaton fire	USA (Los Angeles, California)
Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants	On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants	*Atmos Pollut Res*	Metadata only (no abstract)	Light-duty vehicle on-road particulate and gaseous emissions	France (inferred from author affiliation; not stated in record)
Lian firework barium aerosols single-particle SPAMS Guangzhou	Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring	*Atmos Chem Phys*	Field measurement	Online Ba mass concentration plus single-particle aerosol mass spectrometry (SPAMS)	China (Guangzhou, four sites)
Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk	Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ metals, soluble fractions	Brazil (Rio de Janeiro state)
Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific	Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific	*Atmos Res*	Metadata only (no abstract)	Aerosol chemical composition and light extinction (IMPROVE-type apportionment)	Taiwan (inferred from author affiliation; not stated in record)
Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025	Long-Term Trends and Nonlinear $PM_{2.5}$-O3 Coupling in a Typical Oilfield City of the "2 + 26" Region	*Atmosphere*	Statistical model	Daily $PM_{2.5}$ and MDA8 O3 (fixed monitors), 2015-2025	China (Puyang, Henan)
Other clinical endpoints					
Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China	Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China	*Environ Res*	Cross-sectional survey	Three-year mean landscape fire-sourced $PM_{2.5}$ at school address (modelled)	China (national surveillance)
Hu air pollution built environment lung cancer recurrence after surgery China	Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities	*Research Square (preprint)*	Retrospective cohort	Satellite-derived 1 km $PM_{2.5}$, PM_{10}, NO2, CO, SO2, O3 (post-diagnosis first year)	China (three cities)

Provenance and method

Entry window **28 October 2026**, single day (Wednesday). Harvester legs (run 30 Oct 00:21 UTC): **PubMed** health **16**, sensing **2**; **Europe PMC 26**; **Crossref by ISSN 78**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429. PubMed connector (EDAT 23–28 Oct, bucketed by entrez date) 18 PMIDs for 28 Oct; **118 screened**. Abstract retry (efetch; Europe PMC → Semantic Scholar) recovered none of the 7 day-0 deposits. **22 admitted**, **96 rejected** with logged reasons, of which 14 are cross-day duplicates issued under 27 October. **DOI gate** run before build. Consensus sweep and trial watch for 23–29 October are reported in the 29 October issue. **7 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.